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PSYCH 301
Cognitive Processes

Lecture 12:
Short-Term and Working Memory

Fall 2026
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Working Memory

- Modal model view of STM turned out to be too simple
  - STM is not just passive storage, but also requires “control processes,” so we need a model that includes those somehow…
- Baddeley and Hitch (1974): Working memory - Similar concept to short-term memory
- Working memory: limited capacity system for temporary storage and manipulation of information for complex tasks such as comprehension, learning, and reasoning



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Working Memory

- “multiply 43 times 6 in your head”

  1. Visualize: 43 * 6
  2. Multiply 3 * 6 = 18
  3. Hold 8 in memory, while carrying the 1 over the 4
  4. Multiply 6 * 4 = 24, then add the carried 1 = 25
  5. Combine 25 and the 8 you were holding = 258

  This operation involves both storage (holding 8 in memory) & active processes (carrying the 1…)

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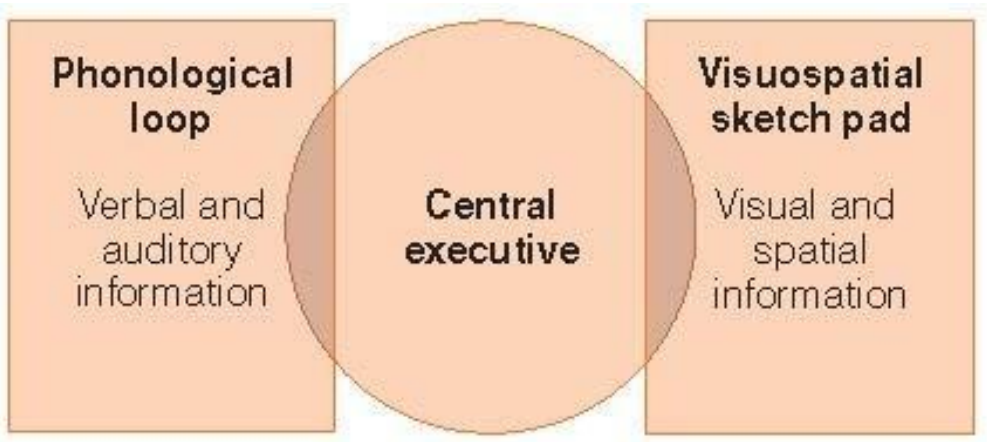
Comparing Memory Models

- Short-term Memory (Atkinson & Shiffrin)
  - Single component for all types of info
  - Mainly used for holding information for a short time
- Working Memory (Baddeley & Hitch)
  - Three components:
    - **Central Executive**
    - **Visuospatial Sketchpad**
    - **Phonological Loop**
    - *(+ Episodic Buffer, added by Baddeley in 2000)*
  - Used for manipulation of information during complex cognition

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Components of Working Memory

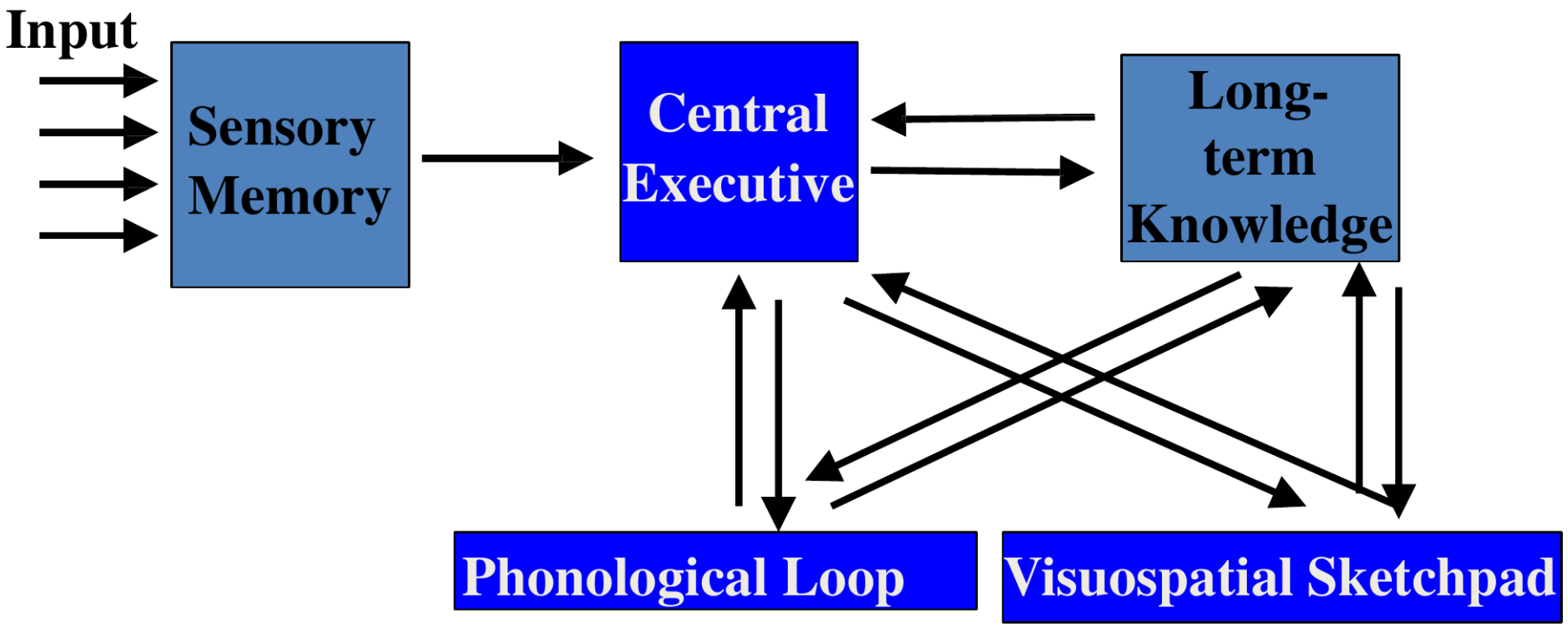
- Phonological Loop
  - Phonological store: Holds verbal and auditory information
  - Articulatory rehearsal process: responsible for rehearsal that can keep items in the phonological store from decaying
- Visuospatial Sketch Pad
  - Holds visual and spatial information
- Central Executive
  - Pulls info from long-term memory, coordinates other components, directs and maintains attention…



Baddeley's working memory model

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Working Memory
Baddeley & Hitch (1974)



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Phonological Loop

- Phonological similarity effect
  - Letters or words that sound similar are confused
- Word-length effect
  - Memory for lists of words is better for short words than for long words
  - Takes longer to rehearse long words and to produce them during recall
- Articulatory suppression
  - Speaking interferes with rehearsal, therefore memory is reduced

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Phonological Similarity Effect

- The basic effect: words that sound similar are confused by people

- Conrad (1964)
  - Participants saw target letters (quickly flashed)
  - Half of the time the letters were similar and half of the time they weren't
  - Then they wrote them down
  - Mistakes were made
    - Not likely to replace with something that looked like the target (E for F)
    - Likely to replace with something that SOUNDED like the target (E for B)

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Phonological Similarity Effect

- The Phonological Similarity Effect experiment:
  - Demonstrated the phonological similarity effect (people confuse letters that sound similar)
    - *even though information was presented visually*
  - As we'll see later, it also showed that the phonological loop is necessary for the conversion (not just holding info)

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Word-Length Effect

- The basic effect: When memorizing words, you can remember fewer words if the words are long
  - People remembered more of the short words than long words (Baddeley et al., 1975, 1984)
  - Welsh–English bilingual children have a shorter digit span in Welsh than in English (Ellis & Hennelly, 1980)
    - Because Welsh numbers take longer to pronounce!
  - The number of words you can say in 1.5-2.0 seconds is likely to be your digit span



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Articulatory Suppression

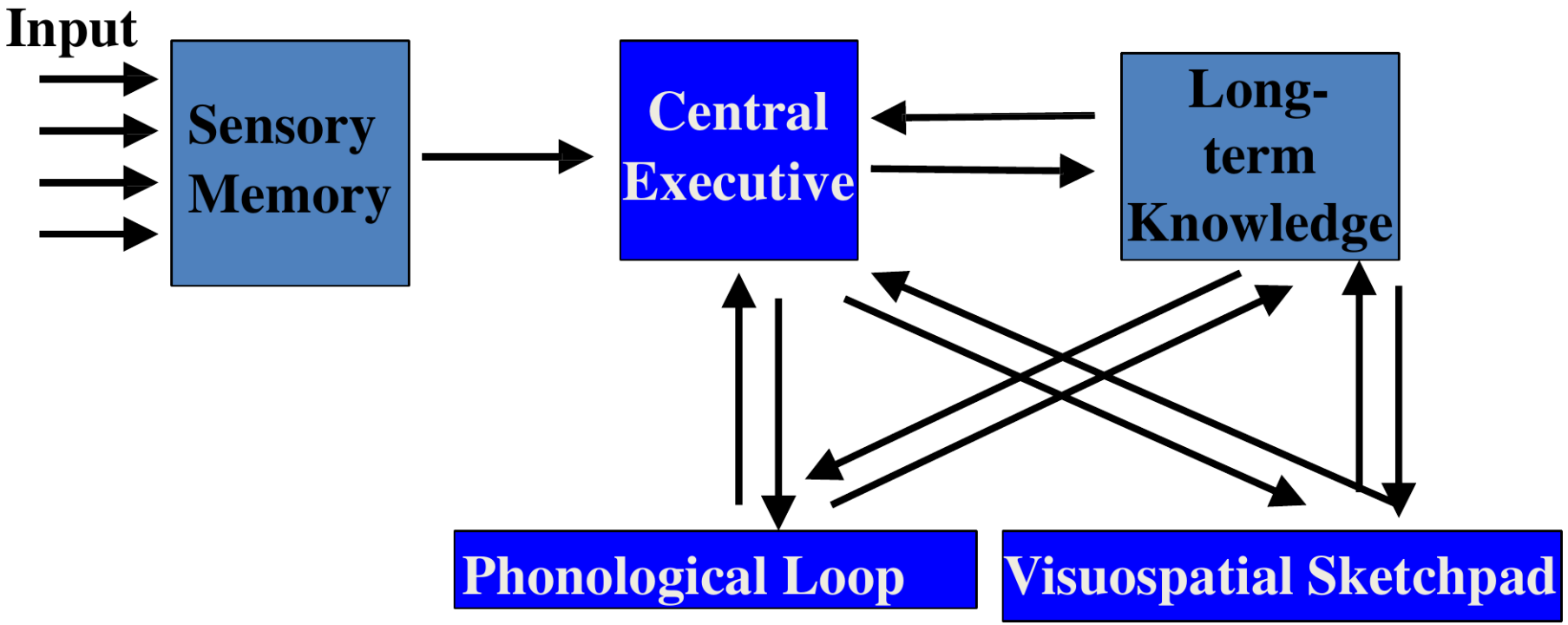
- The basic finding: if you speak while memorizing (which keeps the phonological loop busy):
  - you get worse at remembering,
  - the word length and phonological similarity effects are reduced

- What is going on in these situations? Why does this support the concept of a phonological loop?

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Visuospatial Sketch Pad

- The visuospatial sketch pad holds visual and spatial information
- Involved in the process of visual imagery



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Visuospatial Sketch Pad

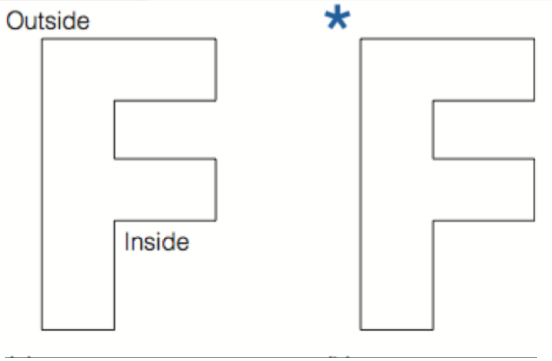
- Brooks (1968) – the sentence experiment
  - Memorize a sentence
  - Indicate whether each word is / is not a noun
    - Condition 1: indicate by speaking
    - Condition 2: indicate by pointing
  - Results: pointing was easier than speaking for the participants

- Explanation: Phonological loop was busy processing the sentence, but sketch pad was free

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Visuospatial Sketch Pad

- Brooks (1968) – the “F” demo
  - Memorize a shape (in this case an F)
  - Indicate whether each corner is an “inside corner” or an “outside corner”
    - Condition 1: Indicate by speaking
    - Condition 2: Indicate by pointing
  - Results: Speaking is easier than pointing (the OPPOSITE of what happened before)
- Explanation: Sketch Pad was busy with image, but phonological loop was free



CORNER	POINT	
1	OUT	IN
2	OUT	IN
3	OUT	IN
4	OUT	IN
5	OUT	IN
6	OUT	IN
7	OUT	IN
8	OUT	IN
9	OUT	IN
10	OUT	IN

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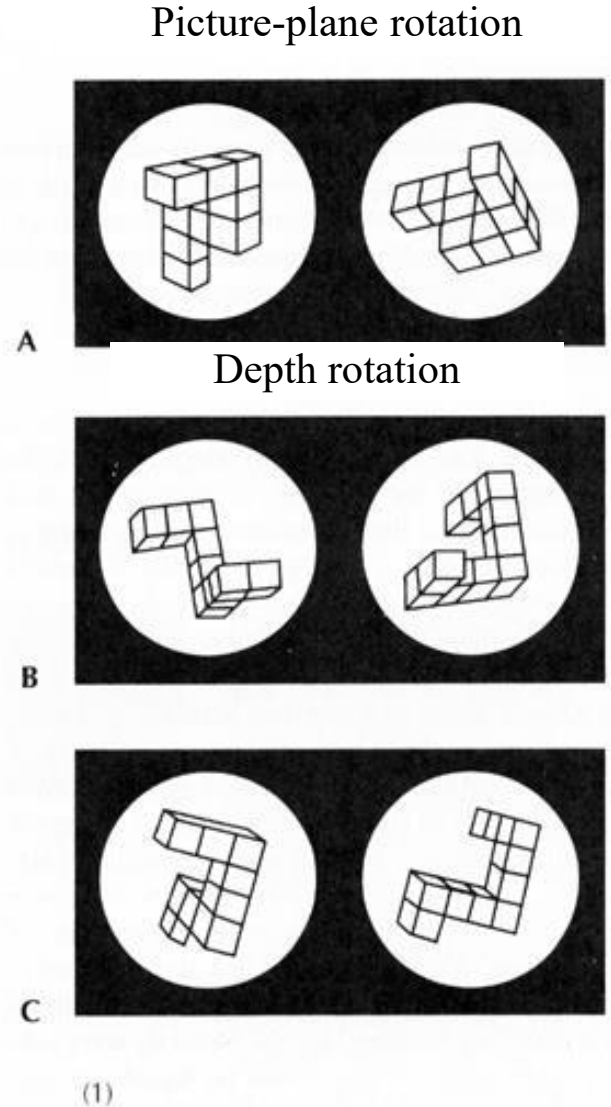
Visuospatial Sketch Pad

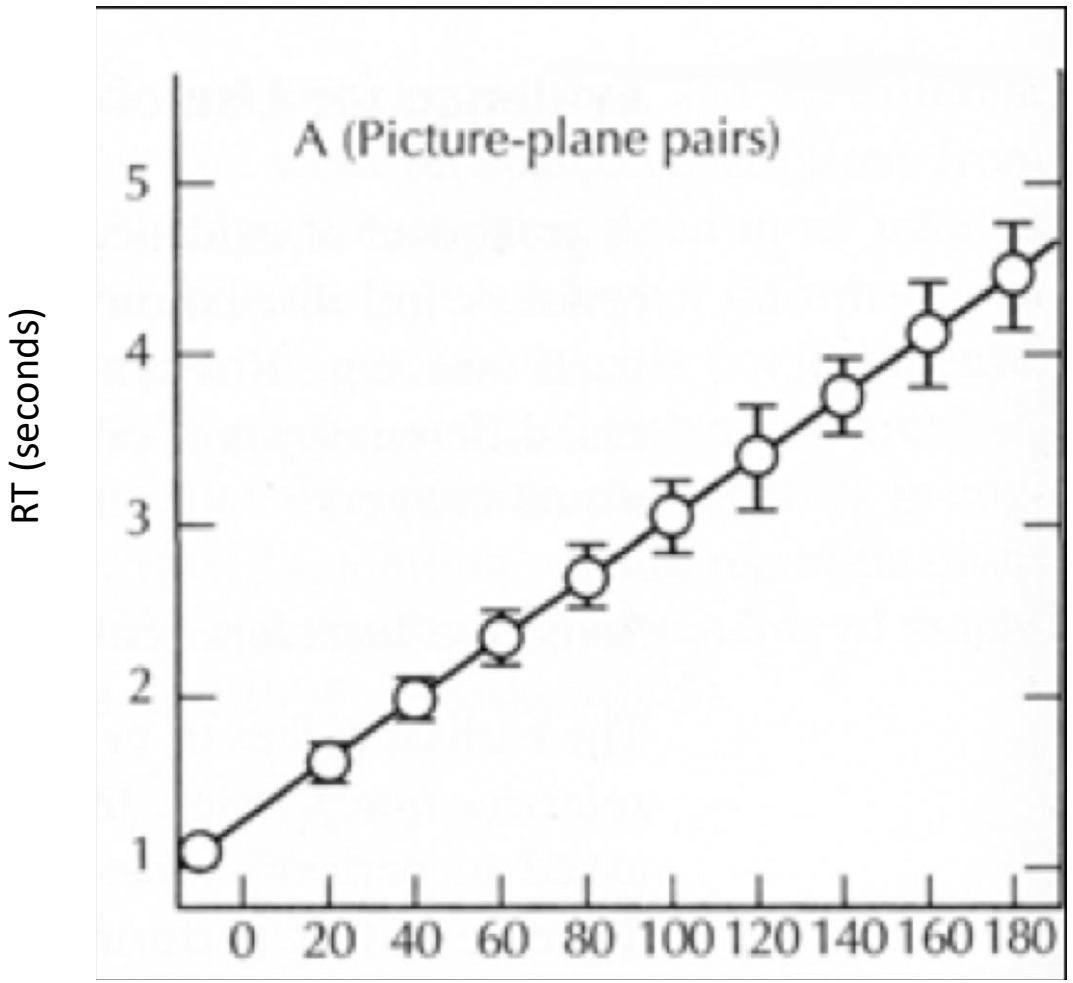
- Tasks are easier when the information being held in mind and the operation being performed on it involve *different* types of short-term memory
  - Verbal / Phonological
  - Visual / Spatial
- That means that the two types of short-term memory are somewhat independent
  - At the least, separate **capacities**

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Mental rotation…

Mental Rotation





Shepard & Metzler (1971)

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The Central Executive

- The Central Executive does the “work” of working memory
  - Directing and maintaining attention
    - Focus, divide, switch attention
  - Controls suppression of irrelevant information
  - Coordinating sketchpad and phonological loop
  - Performing calculations
  - Today often split into updating, shifting, and inhibition (Miyake et al., 2000)

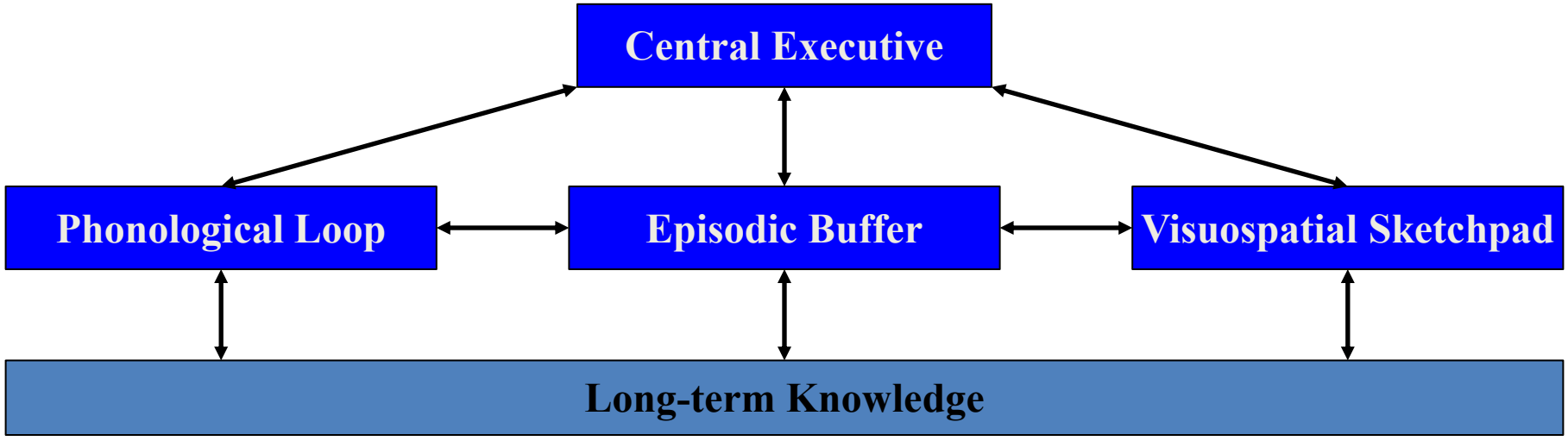


The idea of a “central executive” is often criticized as theorizing a “homunculus” in the brain.

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The Episodic Buffer (Baddeley, 2000)

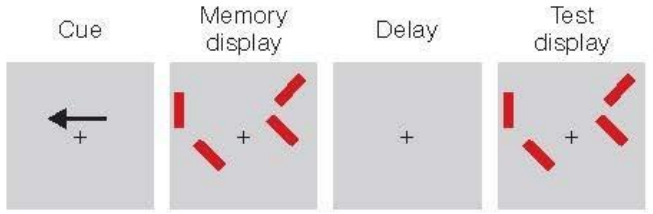
- A fourth component, added to the model in 2000
  - Limited capacity (about 4 “chunks”)
  - Binds information from the loop, sketch pad, and long-term memory into integrated “episodes”
- Why was it needed?
  - People can recall ~16 words of a meaningful sentence, far beyond span for unrelated words
  - Some amnesic patients can immediately recall a just-heard story



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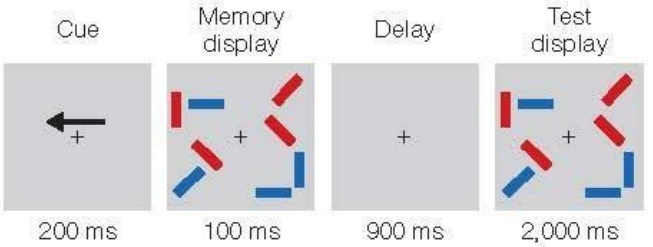
WM and the Brain: Individual Differences

- Vogel et al. (2005)
- Determined participants' WM capacity
  - High-capacity WM group
  - Low-capacity WM group
- Shown either simple or complex stimuli
- Measured ERP responses



(a)

- https://www.youtube.com/watch?v=I3j2VrhqTAA



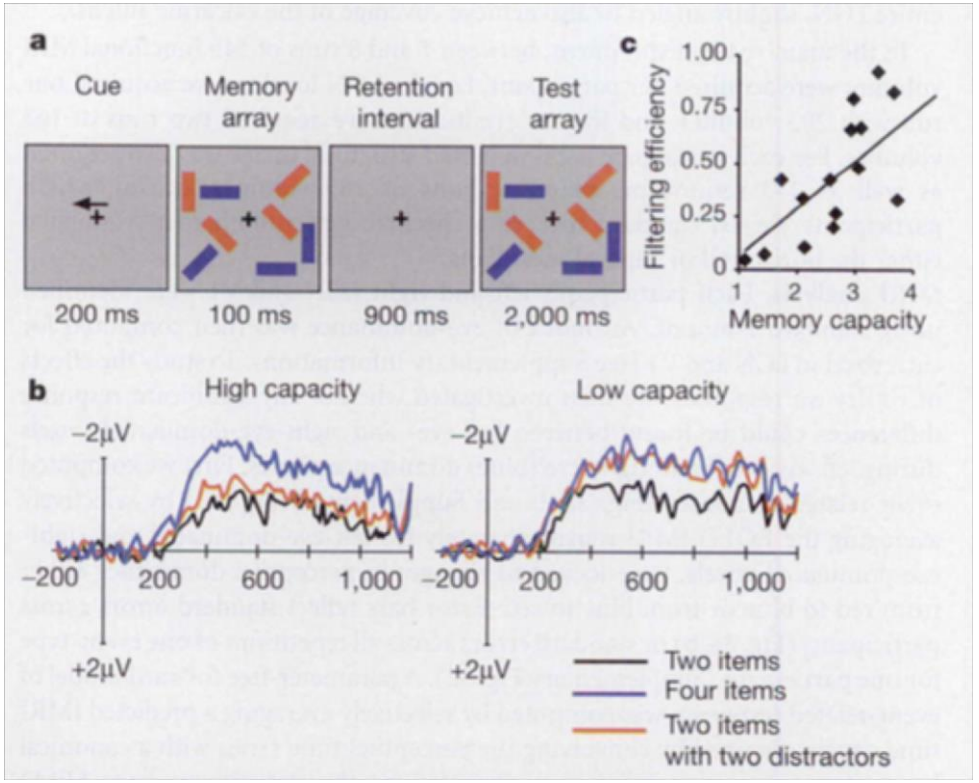
(c)

Consider panel b:

Activity continues well into the "Retention interval."

For High Capacity (HC) individuals, activity is greater when there are four *relevant* items vs. two relevant items.

Adding irrelevant (blue) items has no effect for HC people but increases activity for people with low capacity.



Now consider panel c:

“Capacity” is correlated with the EEG measure of “filtering efficiency.”

People with higher capacity are also less distracted by the irrelevant items.

What does this mean for how we understand differences in working memory capacity?

